

MD. RONI KHAN

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Google Scholar ID: [NImPrLwAAAAJ](https://scholar.google.com/citations?hl=en&user=NImPrLwAAAAJ)

EDUCATION

B.Sc. Engineering in Electrical and Electronic Engineering

Gopalganj-8105, Bangladesh

Passing year: October 2025

Gopalganj Science and Technology University

CGPA: **3.89/4.00** (Ranked first in the batch)

RESEARCH INTERESTS

Deep Learning and Machine Learning; Biomedical Imaging; Quantum Machine Learning; Power-System Dynamics and Stability

PUBLICATIONS

1. **Md. R. Khan**, U. Sutradhar, P. Sarkar, Md. I. Al Imran, and S. A. Dipa, “Deep Learning-Driven Brain Tumor Sectionalisation: Leveraging YOLO11 for MRI Analysis,” in *2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)*, IEEE, Jul. 2025, pp. 1–6. doi: 10.1109/QPAIN66474.2025.11172256.
2. P. Biswas, U. Sutradhar, **Md. R. Khan**, S. Sana, S. Islam, and S. Paul, “YOLO11 for High Accuracy Real-Time Detection and Classification of Diverse E-Waste Categories: Enhancing Recycling Efficiency,” *Engineering Reports*, vol. 7, no. 11, Nov. 2025, doi: 10.1002/eng2.70459.
3. **Md. R. Khan**, A. M. Avi, and S. A. Dipa, “Assessing Two-Stage and Set-Based Detectors for Cervical Cytology,” *2025 IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON)*, pp. 191–196, Nov. 2025, doi: 10.1109/BECITHCON69222.2025.11504190.
4. **R. Khan**, A. M. Avi, S. Mondal, and S. A. Dipa, “Breast Cancer Histology Image Classification using Transfer Learning and Grad-CAM-Based Explainability,” *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, pp. 1–6, Apr. 2026, doi: 10.1109/QPAIN69676.2026.11545899.
5. M. Nayem, P. Biswas, M. J. Naime, **Md. R. Khan**, U. Sutradhar, and Md. N. Uddin, “From Symptoms to Clinical Insight: An Explainable Machine Learning Framework for Early Diabetes Screening,” *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, pp. 1–6, Apr. 2026, doi: 10.1109/QPAIN69676.2026.11546646.

RESEARCH EXPERIENCE

UNDERGRADUATE THESIS

FarmChat: LLM-RAG-Powered Framework for Real-Time Tomato Leaf Disease Detection and Farmer Assistance

Supervisor: Dr. A.T.M. Saiful Islam

- Developed an end-to-end AI framework for real-time tomato leaf disease detection using the YOLO11 object detection model.
- Integrated **Large Language Models (LLMs)** with **Retrieval-Augmented Generation (RAG)** to deliver bilingual (Bangla–English) agricultural decision support for farmers.
- Designed an explainable AI-assisted pipeline to improve the reliability and interpretability of disease diagnosis and recommendations.
- Developed a prototype chatbot that provides disease diagnosis, cultivation guidance, and treatment recommendations through natural language interaction.
- Evaluated the proposed framework using publicly available plant disease datasets, demonstrating its potential for practical precision agriculture applications.

TECHNICAL COMPETENCIES

- **Programming Languages:** Python, MATLAB, R, C, C++
- **Quantum Computing:** Qiskit, PennyLane
- **Power-System Analysis:** MATLAB/Simulink, MATPOWER, Power System Toolbox
- **Deep Learning Frameworks:** PyTorch, TensorFlow, Keras
- **Computer Vision & AI:** OpenCV, YOLO, Detectron2
- **Data Science & Machine Learning:** Scikit-learn, NumPy, Pandas, Matplotlib
- **Development Tools & Platforms:** Git, Google Colab, VS Code, PyCharm, LaTeX
- **Simulation & Engineering Tools:** Proteus, AutoCAD

TRAINING

Industrial Training

Training Institute for Chemical Industries (TICI), Bangladesh

January 2025

- Gained practical knowledge of electrical power systems and operations

LANGUAGES

- Bangla: Native
- English: IELTS Overall **6.5** (Listening: 6.5 | Reading: 6.5 | Writing: 6.0 | Speaking: 6.0)

REFERENCES

Dr A.T.M. Saiful Islam

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Dr Mohammad Asaduzzaman Khan

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